Artificial Intelligence –
Deep learning 5.1: ***Sequence models (Extra material)***

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LSTM: inside the cell

- **LSTM networks** - Long short-term memory cells
  - The most widely used RNN cell
  - More sophisticated hidden function than the basic RNN
  - Capable of learning long-term dependencies

$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$

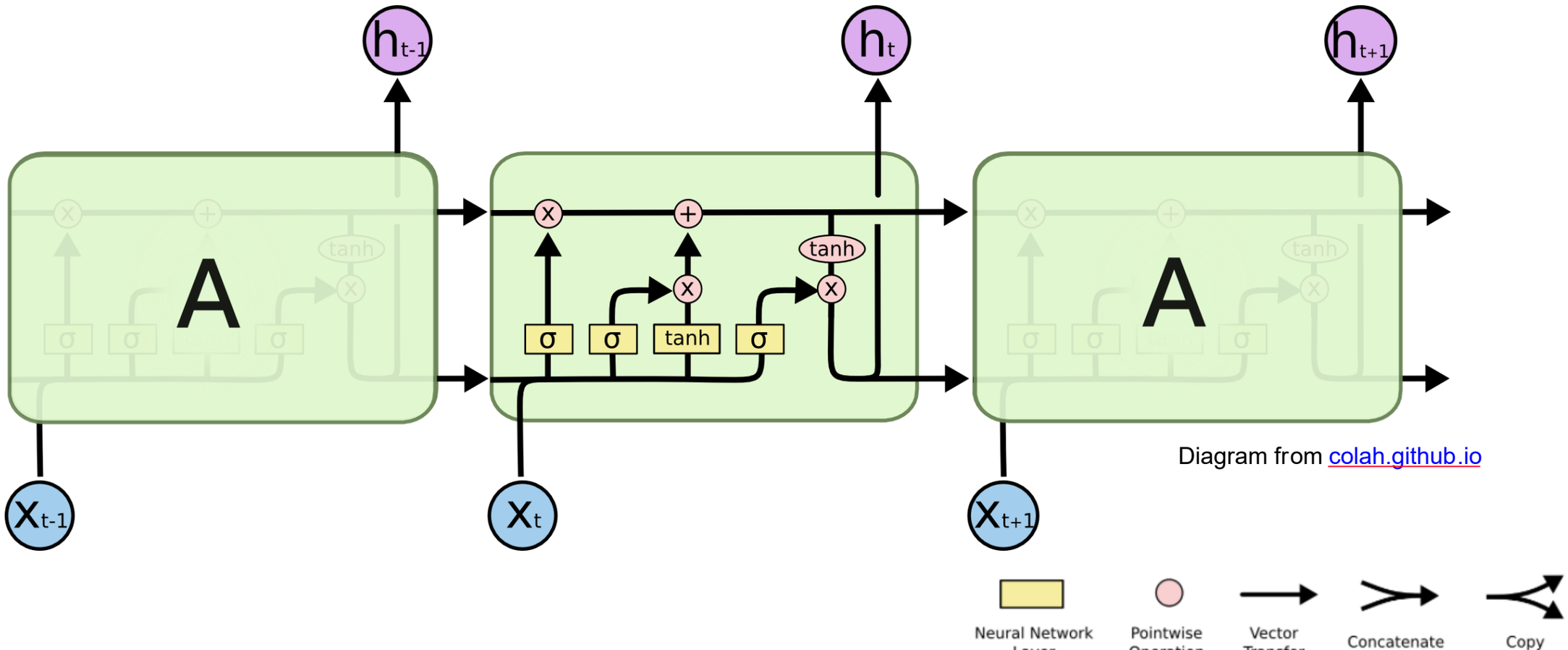
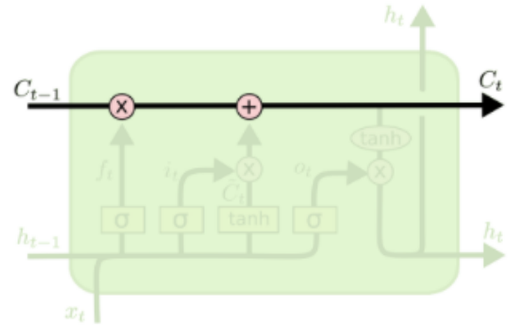
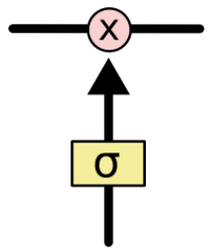


Diagram from colah.github.io

LSTM: inside the cell

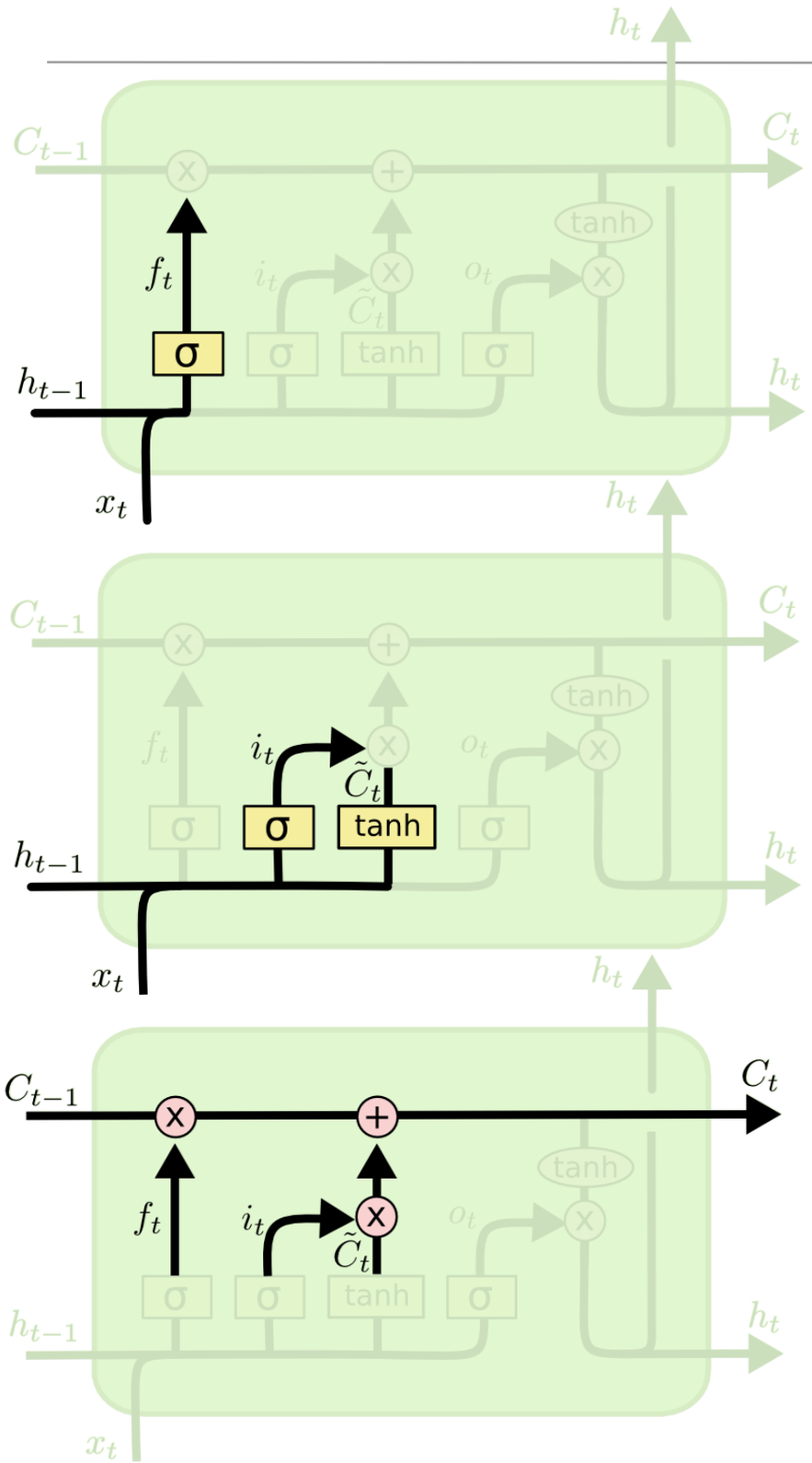




$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$



$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] \; + \; b_f\right)$$

$$i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] \; + \; b_i\right)$$

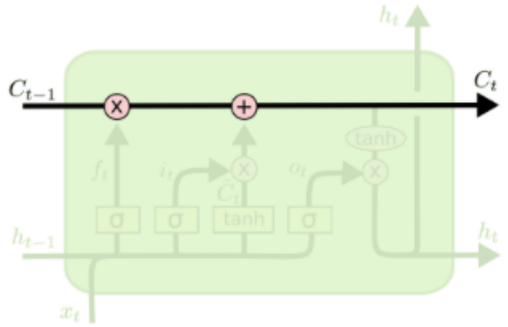
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] \; + \; b_C)$$

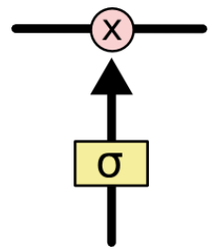
$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

Diagram from colah.github.io



LSTM: inside the cell

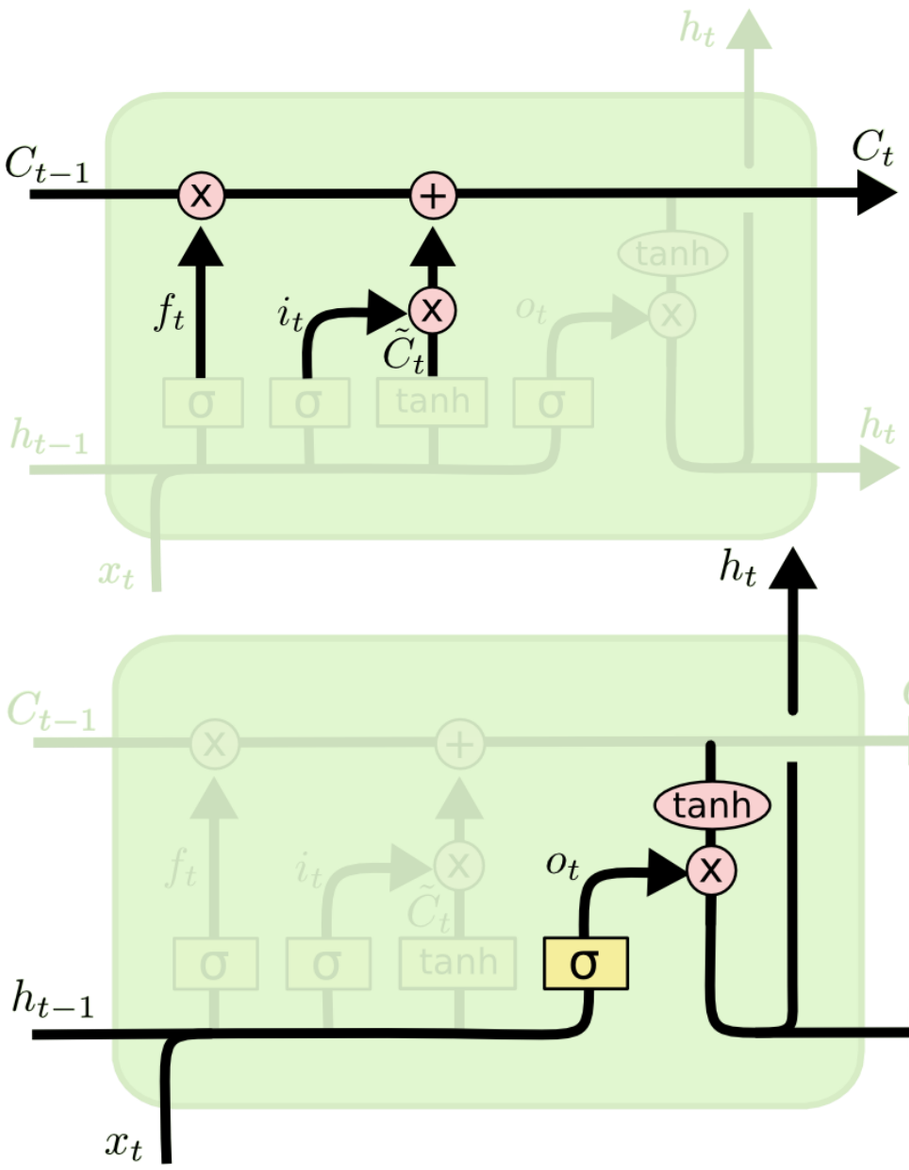




$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$



$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

$$o_t = \sigma\left(W_o\left[h_{t-1}, x_t\right] + b_o\right)$$

$$h_t = o_t * \tanh\left(C_t\right)$$

Diagram from colah.github.io



Exercise: a tiny LSTM by hand

Vocabulary V = 3: a, b, c
Hidden units H = 2 (size of c and h)

$$f_t = \sigma(W_{xf}\, x_t + W_{hf}\, h_{t-1} + b_f)$$
$$i_t = \sigma(W_{xi}\, x_t + W_{hi}\, h_{t-1} + b_i)$$
$$g_t = \tanh(W_{xg}\, x_t + W_{hg}\, h_{t-1} + b_g)$$
$$o_t = \sigma(W_{xo}\, x_t + W_{ho}\, h_{t-1} + b_o)$$
$$c_t = f_t \odot c_{t-1} + i_t \odot g_t$$
$$h_t = o_t \odot \tanh(c_t)$$
$$\hat{y}_t = \mathrm{softmax}(W_y\, h_t + b_y)$$

Part A **Sizes and roles: fill in the table**

object	size	role (one line)
x_t		
W_xf		
W_hf		
b_f		
W_x (4 gates)		
W_h (4 gates)		
c_t		
h_t		
W_y, b_y		
LSTM parameters		

Work in pairs, 5 minutes. Parts B and C on the next slides.

Exercise: what kind of representation?

Part B Same tiny LSTM: V = 3 (a, b, c), H = 2

1. While the network reads one sequence, which of these change: the weights, c_t, h_t?

2. What does the network “know” about the letter b, and where is it stored? How many numbers?

3. True or false: c_t has one entry per letter.

4. We add a fourth letter, d. Which sizes change? Do c and h change size?

5. We double H to 4. Which sizes change, and how many LSTM parameters are there now?

Hint: separate what is learned (parameters) from what is computed while reading (state).

Exercise: one update step by hand

Part C **Weights (already trained)**

		W_x (letters)			*W_h*		
gate	**unit**	**a**	**b**	**c**	**h_1**	**h_2**	**bias**
f	1	1	2	−1	4	0	0
	2	3	−4	0	0	2	0
i	1	0	−4	2	0	1	0
	2	1	4	−2	0	0	0
g	1	1	0	−1	0	0	0
	2	0	2	1	−2	0	0
o	1	2	0	1	0	3	0
	2	0	0	−1	2	0	−1

Previous state

c(t−1) = [1.0, 0.5] h(t−1) = [0.5, 0]

Input letter: b → x_t = [0, 1, 0]

σ: −4 → 0.02, 0 → 0.5, 4 → 0.98
tanh: 0 → 0, 0.76 → 0.64, 0.98 → 0.75, 1 → 0.76

1. Which column of W_x is used? Compute a_f, a_i, a_g, a_o.
2. Apply σ / tanh: f, i, g, o.
3. Compute c_t and h_t.
4. What did each unit do: keep, erase, write?
5. Bonus: W_y = [[1, 0], [0, 1], [−1, −1]], b_y = 0. Next letter?

Uses the equations from Part A. 10 minutes.